How to access the ICE cluster and run the EEG notebooks

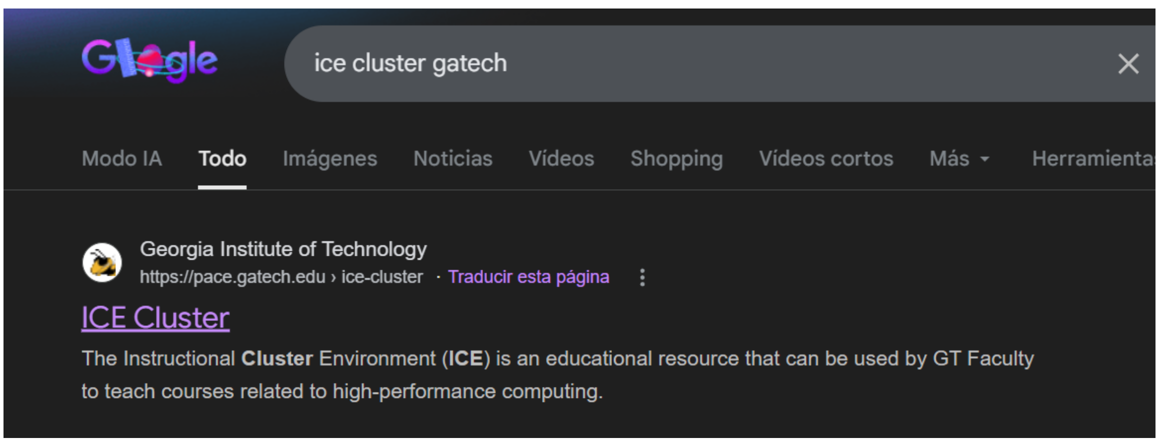
This tutorial explains how to open a VS Code session on the ICE cluster and set up the environment so you can run the EEG quickstart and advanced notebooks (*utils.py, student_eeg_quickstart.ipynb, student_eeg_advanced.ipynb*) and the subsequent analysis.

You will need your Georgia Tech login credentials, and you already have access to the ICE cluster.

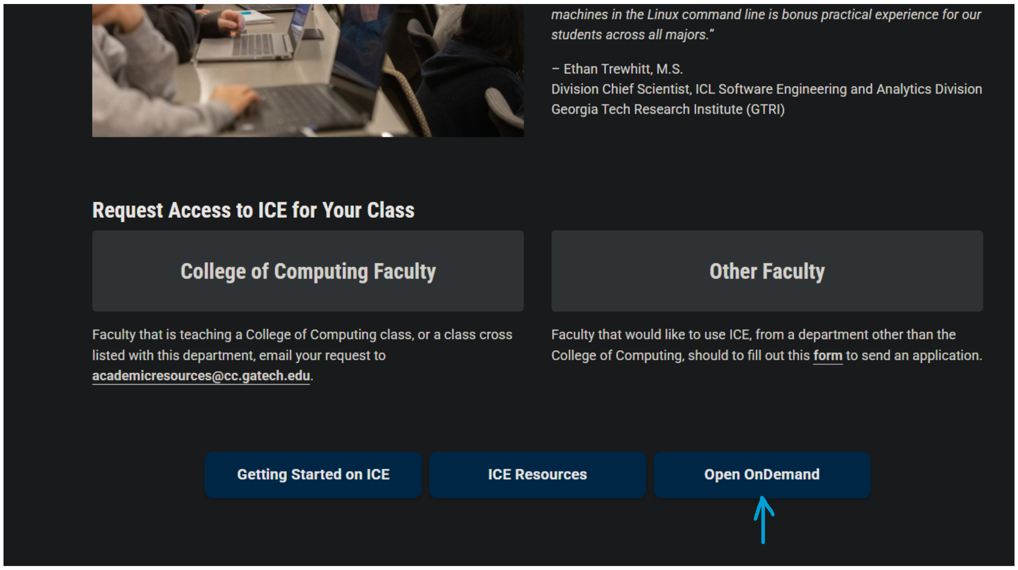
Step 1: Log in to the ICE portal

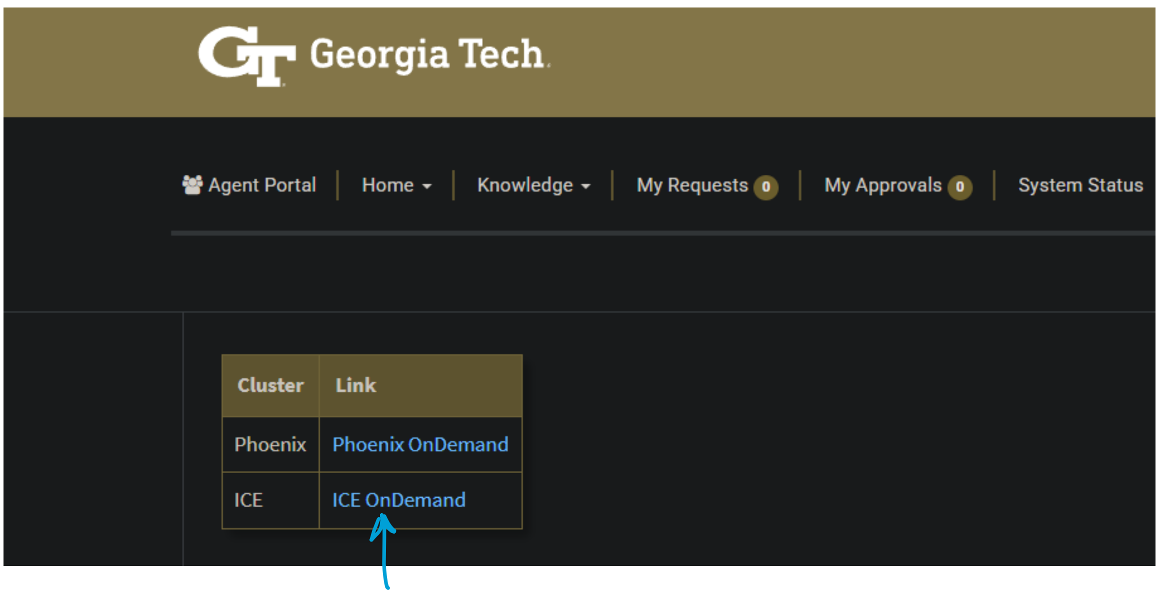
Open the ICE / PACE portal in your browser and log in with your GT credentials.

https://pace.gatech.edu/ice-cluster/



Scroll down and press "Open OnDemand"

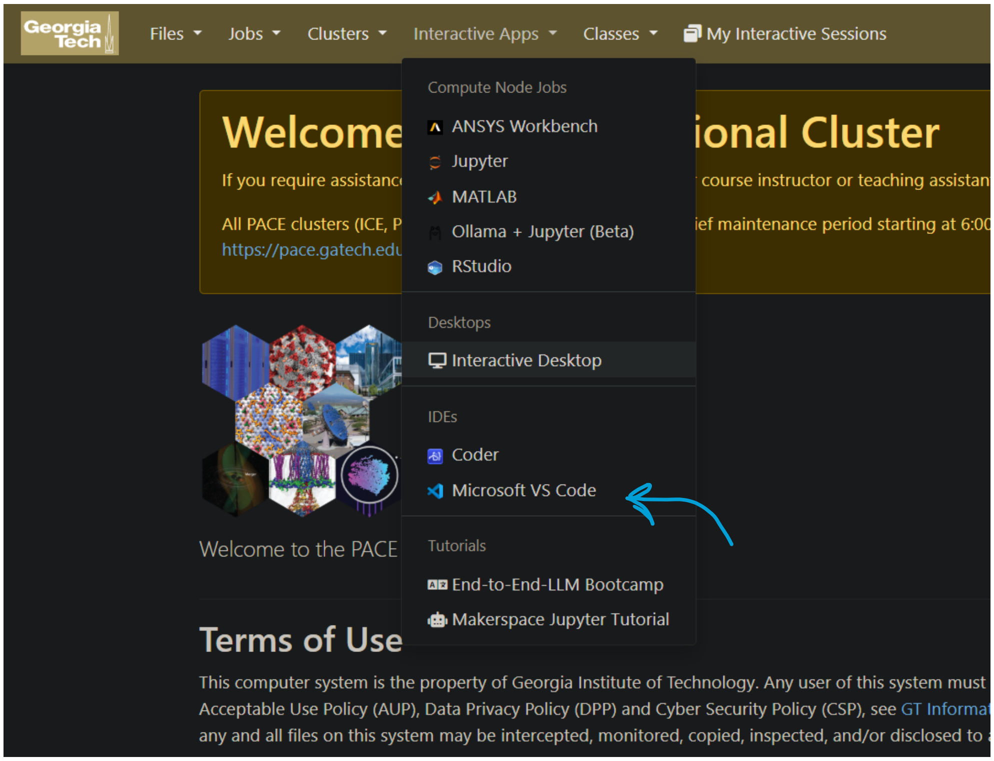




You will need to use the Georgia Tech VPN if you are not on campus.

Step 2: Open Interactive Apps and start VS Code

In the left menu, go to Interactive Apps. Under IDEs (or Compute Node Jobs), click Microsoft VS Code. You may see a form to configure your session.



Step 3: Configure the session

On the form, use the following exact settings to ensure you have enough power for EEG processing:

- **Quality of Service:** Select pace-ice (or the one assigned to your specific course like coc-ice).
- **Node Type:** Select CPU only, first available.
- **Number of cores (CPUs):** Enter 4.
- **Memory (GB):** Enter 16.
- **Number of hours:** Enter 1 or 2.

To ensure MNE and Jupyter work correctly, you must not use the "Default" setup.

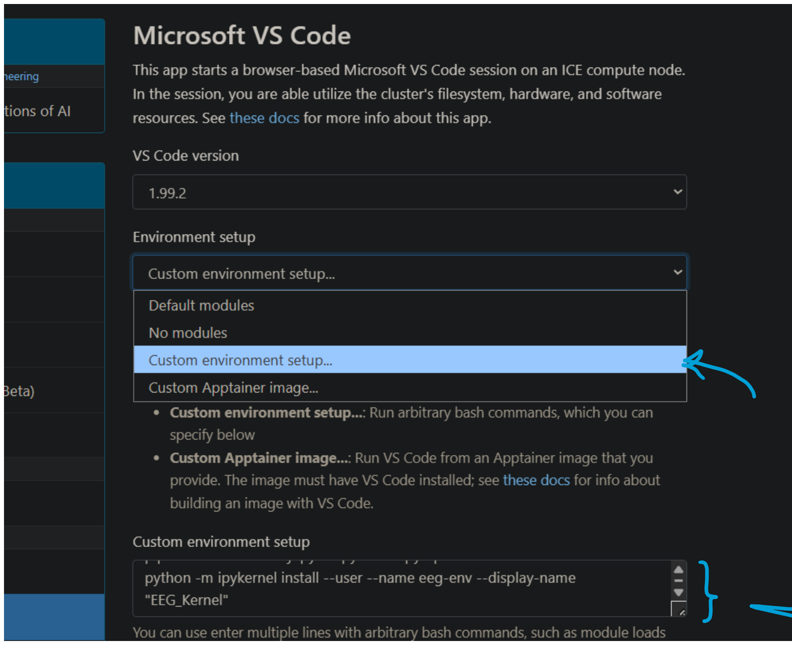
1. In the **Environment setup** section, select **Custom environment setup...**.
2. In the text box that appears, paste these exact lines:

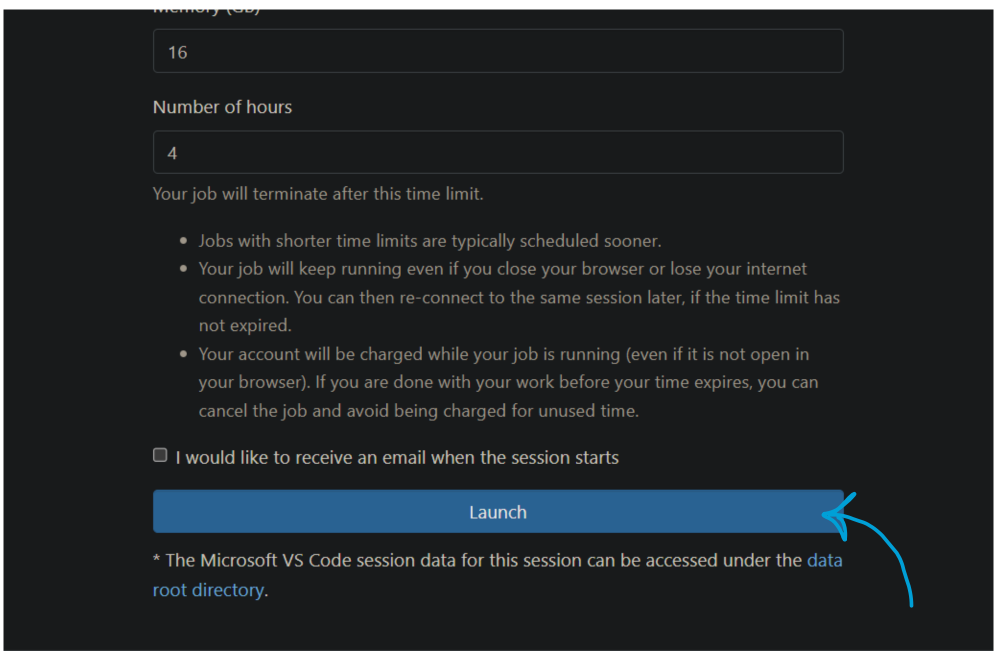
module load miniforge/24.3.0-0

mamba activate base

pip install --user mne jupyter ipykernel ipympl python -m ipykernel install --user --name eeg-env --display-name "EEG_Kernel"

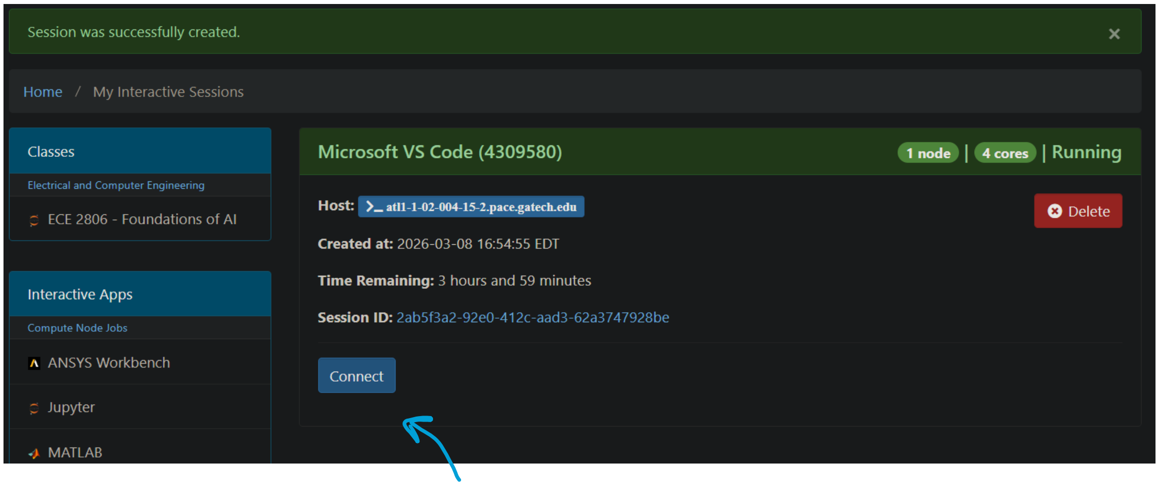
3. Click **Launch** (or Submit) at the bottom of the form.





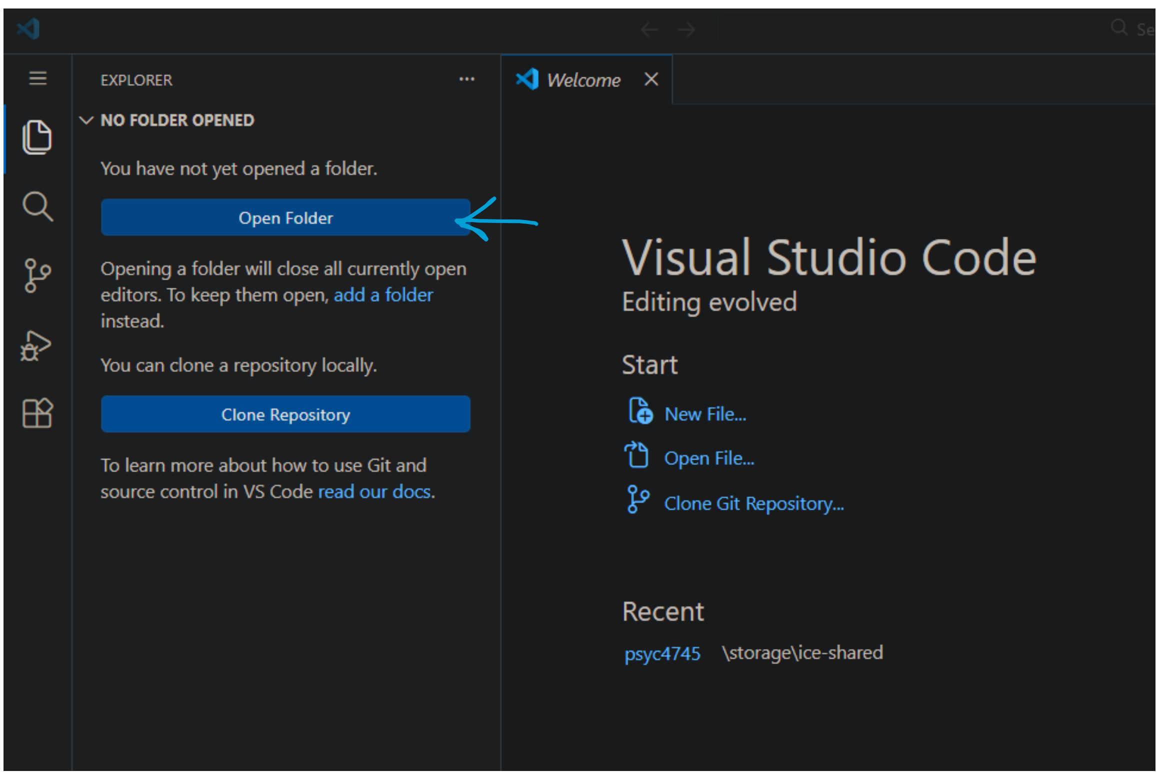
Step 5: Wait for the session and open VS Code

Your job will be queued. When it starts, a link or button to open VS Code in the browser will appear. Click it to open the VS Code session on the compute node.

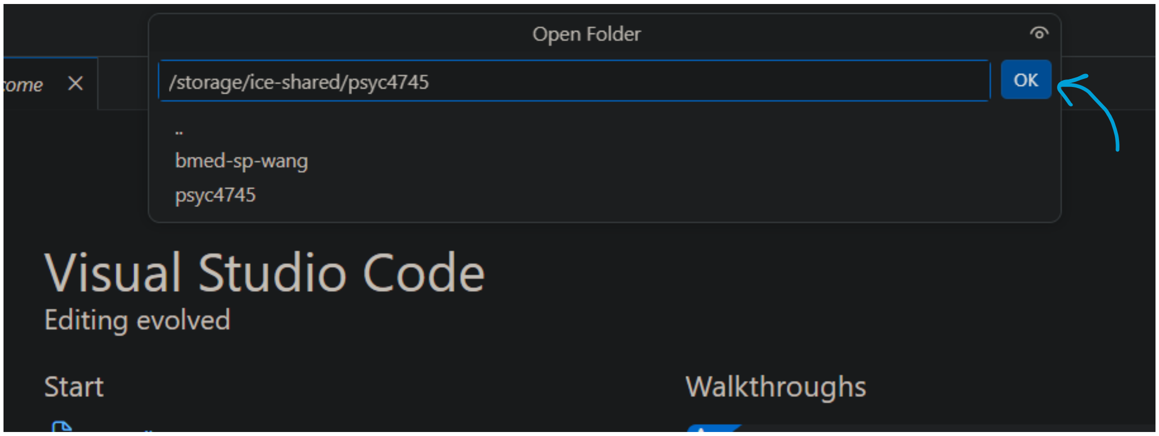


Step 6: Open the project and run the notebooks

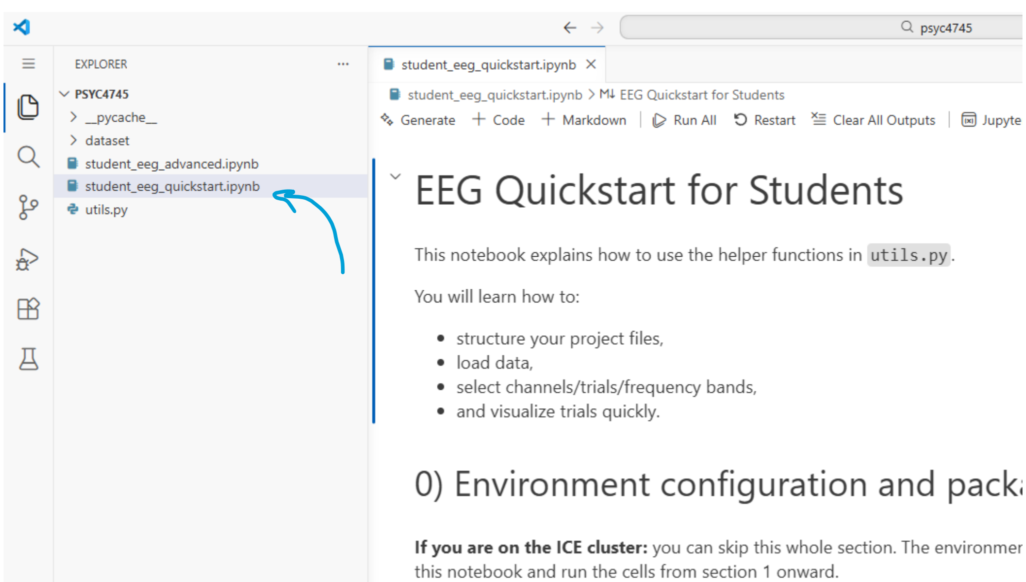
In VS Code, open the folder where the repo is. Open student_eeg_quickstart.ipynb, select the Python kernel (e.g. the one from the environment you activated), and run the cells. You can then open student_eeg_advanced.ipynb the same way.

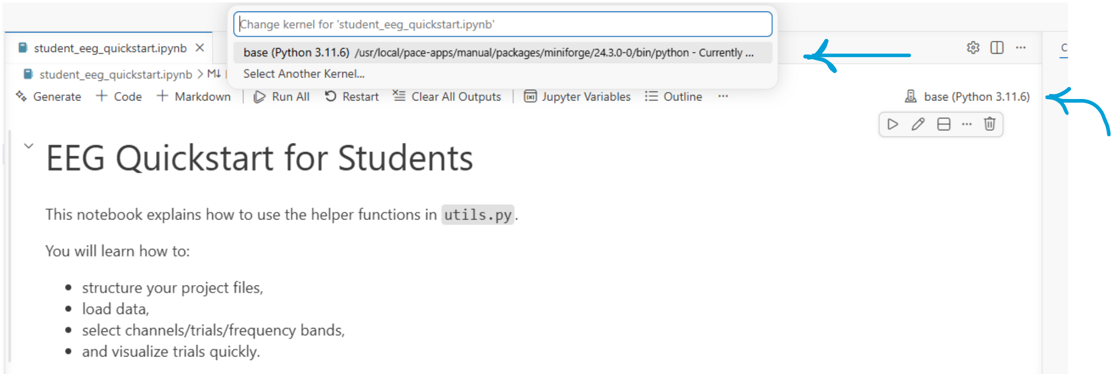


Write the path: “/storage/ice-shared/psyc4745” and press OK.



1. Open your .ipynb file.
2. In the top-right corner, click **Select Kernel** -> **Python Environments...**.
3. Click the **Refresh icon** (circular arrow) and wait 10 seconds.
4. Select the one labeled **"EEG_Kernel"**.
5. If you need interactive plots, ensure the first cell of your notebook contains %matplotlib widget.
6. **IMPORTANT:** When VS Code opens, look for a notification at the bottom right asking to install the **"Python" extension**. Click **Install**.





Troubleshooting: "I can't see the EEG_Kernel"

If the kernel doesn't appear after refreshing:

1. Go to **Terminal -> New Terminal** inside VS Code.
2. Paste this line and hit Enter:

```
module load miniforge/24.3.0-0
mamba activate base
pip install --user mne jupyter ipykernel ipympl
python -m ipykernel install --user --name eeg-env --display-name "EEG_Kernel"
```

3. Refresh the kernel list again.